GROUNDCONTROL

Nodal Series

Targeted Low-Frequency Control for Small Rooms

You dial in the mix. The low end feels tight. The kick hits right. The bass sits perfectly. Then you play it somewhere else, and it falls apart. The low end disappears. Or it overwhelms everything. The balance is gone. So, you come back to your room… and it sounds fine again.

> *You didn't get the mix wrong. Your room did.*

What you're hearing in the low end isn't just your mix, it's your room shaping it.

At bass frequencies, your room dominates what you hear. Certain notes build up and ring. Others cancel out. That "tight" low end you're hearing is being boosted, smeared, or masked by the space you're working in. And because it happens every time you listen, your ears adapt. Your decisions follow. Your mixes follow.

> *If the room isn't controlled, neither is your low end.*

The Problem Lives Where Most Treatment Doesn't

Below ≈120 Hz, your room stops behaving like a space and starts behaving like a resonator.

Bass energy builds between boundaries, reinforcing specific frequencies and sustaining them long after the signal stops. These resonances caused by room modes can ring for over a second, masking detail and distorting balance.

Meanwhile, standard acoustic treatment does very little here.

Foam, panels, and typical "bass traps" work well in the mids and highs. But at low frequencies, they're simply too thin to absorb meaningfully. The result:

- A room that feels controlled… but only above ≈150 Hz
- And a low end that remains unpredictable, inconsistent, and misleading

This Is Why Mixes Don't Translate

In most small studios, the critical problems live between **40 and 100 Hz**—the exact range where:

- Kick and bass define the mix
- Translation issues originate
- Conventional treatment is least effective

Fixing this requires treating the frequencies most rooms can't control. That's what the Nodal Series is designed to do.

> *When the low end is finally controlled, it often sounds less impressive at first — that's how you know it's working.*

Why Your Room Sounds the Way It Does

Bass frequencies behave differently from everything else in a small room. At low frequencies, roughly below 120 Hz, the room itself becomes a musical instrument. Sounds at certain frequencies bounce back and forth between parallel walls and reinforce themselves, building up into sustained resonances called room modes. Some notes ring for a second or more after the music stops. Others are artificially quiet. The result is a frequency response that can vary by 20 dB or more just by moving your head a foot in any direction.

Standard acoustic treatment — foam panels, hanging baffles, corner traps made from thick mineral wool — genuinely works, and you should use it. Above 150–200 Hz porous materials absorb effectively and make a real difference to flutter echo, mid-range coloration, and stereo imaging. If you haven't treated your room at all, start there.

But porous absorption has a hard physical limit in the bass. To absorb sound, a porous material needs to be a meaningful fraction of the sound wavelength in thickness. At 55 Hz the wavelength is over 20 feet. A 4-inch rockwool panel is about 1% of that wavelength — its absorption coefficient at that frequency is essentially zero. To treat 55 Hz with porous material alone you would need panels over 12 inches deep filling most of your corners. Most people don't have the space, and even if they did the mid-range absorption would be excessive.

The result is that most home studios are treated adequately in the mid-range and completely untreated below 80 Hz — the frequency range where the most damaging modal resonances live.

> *In a typical untreated home studio, the room is still ringing audibly at 55 Hz for over a second after the music stops. Your monitors go silent. The room keeps playing.*

22–88 Hz	3 Units	6.5 inches
Continuous Coverage	**40 · 55 · 70 Hz**	**Unit Depth**
the full home studio problem range	*overlapping Q-controlled bands*	*identical across all three units*

A Different Approach

The Nodal Series is designed specifically for this problem. Instead of relying on thickness, it uses tuned resonance, targeting the exact frequencies where small rooms fail. The result is controlled decay, reduced modal ringing, and a low end that reflects what's actually in your mix, not what your room is adding to it.

Each unit is a sealed-cavity Tuned Diaphragmatic Absorber (TDA): a Richlite membrane with compliant bonding to a Baltic birch frame, acting as a decoupled rigid piston. When bass at the target frequency strikes the membrane, it resonates and dissipates acoustic energy as heat through the Owens Corning 705 cavity fill. Tuned absorption. Minimal depth. Corner placement where modal energy is highest.

The Nodal Series brings that engineering to the home studio in a product designed around the frequencies that cause the most problems in typical small rooms and sized to fit in the corners where those problems are worse. It's designed to work alongside your broadband treatment, not instead of it. Porous panels handle everything above 100 Hz. The Nodal Series handles the range below 90 Hz that porous panels cannot reach. The two approaches are complementary, each doing the job the other cannot.

> *The question is not whether to use broadband treatment, you should. The question is whether your specific room has problems below 80 Hz that only a tuned absorber can fix. If it does, few solutions at a practical depth can solve it.*

What That Means in Practice

When the low end is controlled:

- You stop overcorrecting with EQ
- Kick and bass sit naturally in the mix
- Decisions translate outside your room
- What sounds right… is right

Why Corners?

Bass modes build up pressure at room boundaries — the walls, floor, and ceiling. At a corner where two walls and the floor all meet, that pressure is at its absolute maximum. A panel absorber placed in a corner is working at the highest possible concentration of bass energy, making it far more effective than the same unit placed mid-wall. Four units in corners do more than eight units spread around the walls.

Placement matters more for tuned absorbers than for porous panels. A tuned unit works best at the pressure maximum for its target frequency. The corner is the right location for the same reason it is for any bass treatment: it has the greatest energy concentration regardless of room dimensions.

Is the Nodal Series Right for Your Room?

The Nodal Series makes the most compelling case in rooms where the dominant problems are below 80 Hz. Whether that describes your room depends on your specific dimensions. Rooms in the 12 to 24 feet range typically have their worst axial modes clustering between 40 and 100 Hz, right in the Nodal Series range. Larger rooms push the dominant modes lower; smaller rooms push them higher.

If you have taken RT60 measurements with REW and see elevated decay times between 40 and 100 Hz, the Nodal Series is designed for exactly your problem. If your room's worst decay times are above 100 Hz, broadband porous treatment alone may be sufficient and is the right first investment regardless.

The simulation at nodal.integeraudio.com lets you enter your room dimensions and see the predicted RT60 before and after Nodal treatment. It is the most direct way to answer whether your room has a problem the Nodal Series can fix.

> *If your room has persistent problems below 80 Hz, the Nodal Series can address them and nothing else at practical depth can. If your room's problems are above 100 Hz, broadband treatment is the right answer.*

The Products

Nodal 40 — Deep Bass Absorber

The Nodal 40 targets the lowest tuning frequency in the system. Its Richlite membrane is loaded with mass-loaded vinyl (MLV) bonded to the back face, raising the total surface mass density. This additional mass, combined with the air cavity, tunes the piston resonance precisely to 40 Hz. The Q factor of 1.1 delivers the widest absorption bandwidth in the system, extending coverage down to 22 Hz on the lower skirt.

The Owens Corning 705 is positioned behind the membrane — the "forward" position that maximizes viscous coupling to the moving panel face and sets the low Q.

	Nodal 40
Tuning frequency	40 Hz
Q factor	1.1
−3 dB coverage	22–58 Hz
Exterior (W×H×D)	627 × 755 × 164 mm

Nodal 55 — Mid-Bass Absorber

The Nodal 55 uses a Richlite membrane with no MLV. At the common air cavity, bare Richlite resonates at 54.02 Hz, within 2% of the 55 Hz target and well inside the ±25 Hz (−3 dB) bandwidth at Q 1.5. No mass loading is required, and the cabinet depth is identical to the Nodal 40.

The OC 705 is centered behind the membrane — a mid-cavity position that balances viscous coupling and sets the moderate Q 1.5 bandwidth, bridging cleanly between the Nodal 40 below and the Nodal 70 above.

	Nodal 55
Tuning frequency	54 Hz (54.02 Hz verified)
Q factor	1.5
−3 dB coverage	37–73 Hz
Exterior (W×H×D)	627 × 755 × 164 mm

Nodal 70 — Upper Bass Absorber

The Nodal 70 also uses a bare Richlite membrane. To tune to 70 Hz it requires a shorter active air cavity. The exterior cabinet is identical in depth to the Nodal 40 and Nodal 55: a false back panel is glued and sealed flush inside the back panel, reducing the active cavity without changing the exterior footprint.

The OC 705 is placed flush against the false back where the “rear” position minimizes viscous coupling and sets the tightest Q 1.9 in the system. The high Q keeps absorption focused on the dense 70–90 Hz cluster where the Nodal 70’s upper skirt provides coverage alongside broadband porous treatment above 100 Hz.

	Nodal 70
Tuning frequency	70 Hz
Q factor	1.9
−3 dB coverage	52–88 Hz
Exterior (W×H×D)	627 × 755 × 164 mm

What the System Covers

Used together, the Nodal 40, Nodal 55, and Nodal 70 provide continuous overlapping absorption from 22 to 88 Hz with no gaps between tuning points. The three Q-differentiated bands are deliberately staggered: Q 1.1 for maximum reach at the bottom, Q 1.5 bridging the middle, and Q 1.9 focused on the densest room-mode cluster at the top.

Mode Frequency	Unit	Coverage (−3 dB)	RT60: Untreated	RT60: Treated	Reduction
35 Hz	4 x Nodal 40	22–58 Hz	3.30 s	0.61 s	−81%
55 Hz	4 x Nodal 55	37–73 Hz	1.32 s	0.38 s	−71%
70 Hz	2 x Nodal 70	52–88 Hz	0.77 s	0.36 s	−53%
System	4+4+2 units, corners	22–88 Hz	—	—	53–81%

Reference room: 20 × 16 × 8 ft plasterboard walls

> *The target modal decay time for critical listening environments is 0.25–0.55 s. The treated modes in the reference room achieve decay times within or near this range.*

The Corner Stack

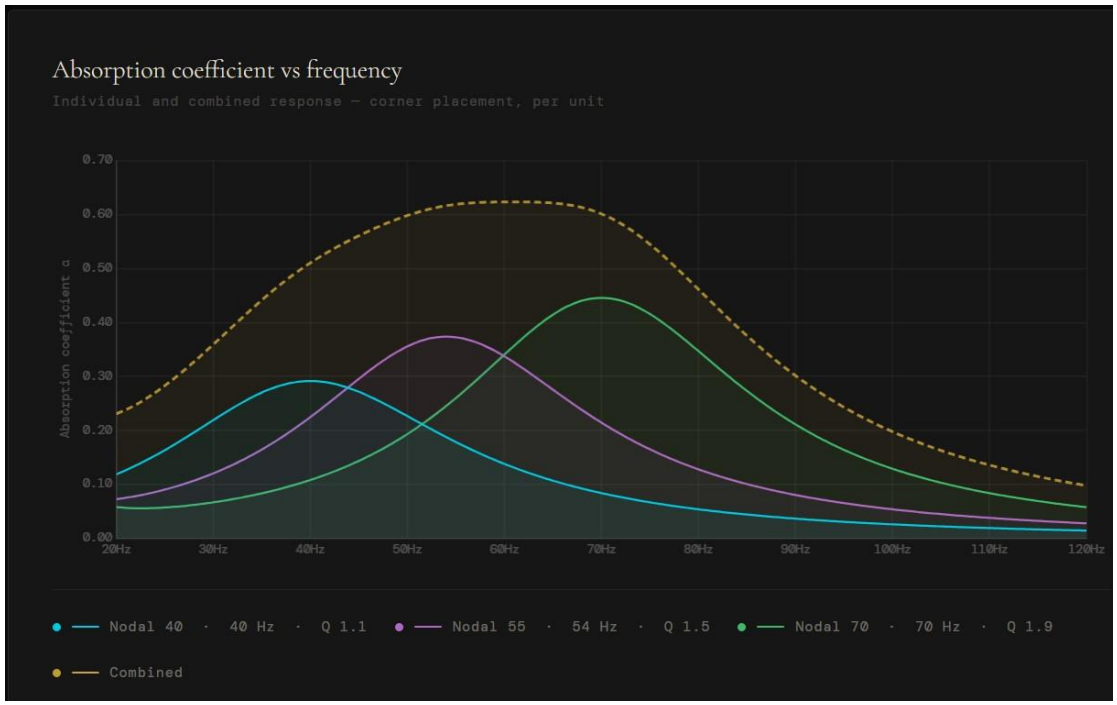
The Nodal Series is designed around a specific installation: three units stacked vertically in each room corner. The Nodal 40 sits at the bottom where sub-bass pressure is highest; the Nodal 55 in the middle; the Nodal 70 at the top. All three units share the same 627 mm wide by 164 mm deep exterior, presenting a single flush column from floor to ~2265 mm, fitting under a standard 8 ft ceiling.

This arrangement is deliberate. The Nodal 40 couples to the floor-corner pressure maximum where the deepest modes concentrate. The Nodal 70 at the top couples into the ceiling corner where higher frequency modes peak. Both pressure maxima are addressed simultaneously by a single flush column that takes up less than 6.5 inches of wall depth.

Complete System Logic

The three frequencies are not arbitrary choices or round numbers selected for convenience. They represent the outcome of working through room mode statistics across more than a thousand room combinations, identifying where membrane absorbers are superior to alternatives, determining the minimum number of units needed to meet the target range with Q-controlled overlapping coverage, and selecting center frequencies at appropriate intervals that maximize coverage of the highest-density mode clusters while ensuring each unit operates where its specific Q value and the membrane absorber mechanism deliver the best possible performance.

The result is a three-unit system that covers approximately 27–88 Hz with continuous overlapping absorption, addresses the most common modal problems across the widest range of real room sizes, and does so using a mechanism that has no practical competition from porous broadband solutions across the entire coverage range.



Concept rendering – AI generated

Starting with Fewer Units

If budget requires a phased approach, start with one Nodal 40 and one Nodal 55 at the two front corners behind your monitors. Front corners couple most strongly to the listening position and give you the greatest immediate improvement. Add the Nodal 70 and rear corner treatment as a second phase.

What to Expect

The first session in a properly treated room often feels unsettling. The low end sounds thinner, less impressive, less present. That's not a problem, it's the point. The resonant reinforcement you were mixing with is gone, and what you're hearing now is closer to what's in the recording.

Give it a few sessions. Once your ears adjust to the treated room, you'll find:

- Bass guitar and kick drum sit in the mix without needing heavy corrective EQ. You're hearing them clearly for the first time
- Low end decisions made in the room translate to other systems. The bass that sounds right actually is right
- The frequency balance at the listening position is more consistent, less variation as you move around the room
- Subwoofer integration becomes more predictable. You can hear what it's actually doing

What the Nodal Series does not address:

- Flutter echo and mid-range coloration. Broadband porous panels handle this above 100 Hz and should be part of any complete treatment
- Every bass mode in your room. Other modes in the bass range may still be audible and may benefit from additional broadband treatment at useful depths
- Rooms where the dominant problems are above 100 Hz. If your RT60 measurements show the worst decay times above that range, porous treatment is the right first investment
- Phase issues or stereo imaging problems. These are separate phenomena unrelated to modal decay

> *The goal isn't a room that sounds impressive. It's a room that tells the truth.*

Key Performance Range

22–88 Hz	3 Units	6.5″ depth
Frequency Range	**40 · 55 · 70 Hz**	**Unit Depth**
continuous coverage	*per corner stack*	*all three units identical*

Finish & Construction

Every element of the Nodal Series is chosen to look as good as it performs. The enclosure is CNC-cut 19 mm Baltic birch plywood — precise, stable, and distinctive. The Richlite membrane panels are finished with Hardwax Oil to an obsidian matte black. The contrast between the natural ply frame and the dark membrane face is the product's visual identity: a warm birch border surrounding each tuned surface, reading as intentional and technical rather than decorative.

About Integer Audio

Integer Audio is a boutique electronics and acoustics company. The Nodal Series is designed and engineered from first principles and honest about what it does and doesn't do.

We build tools for people who take sound seriously.

www.integeraudio.com

User Guide

The Nodal Modal Simulator

Predicting Results Before You Buy

The GroundControl Nodal Series is a precision tool. To ensure you are deploying it effectively, we provide a Modal Simulation tool that calculates the specific resonant behaviour of your room.

Step 1: Define Your Room Dimensions

Enter your room's Length, Width, and Height. The simulator will automatically calculate your Axial Modes, the frequencies where sound waves bounce between parallel surfaces and create "ringing."

- **Pro Tip:** Pro Tip: Look at the Dominant Mode Decay Summary. If you see decay times (RT60) above 0.75 s in the 40–100 Hz range, your room is "Boomy" and requires tuned treatment.

Step 2: Place Your Virtual Nodal Units

The simulator allows you to virtually "install" Nodal 40, Nodal 55, and Nodal 70 units at specific positions.

- **Corner Loading:** Click the corners on the floor plan. The simulator defaults to corner placement because pressure is highest there, making the resonators more efficient.
- **Stacking:** Each unit is independently tuned — Nodal 40, Nodal 55, or Nodal 70. Place each unit separately to build your full corner stack and see the combined effect.

Step 3: Analyze the Heatmap

Toggle between the [Untreated] and [Treated] views.

- **The Heatmap:** Shows "Hot Spots" (High SPL) and "Nulls" (Low SPL).
- **The Goal:** You want to see the "hot" areas (dark reds) become more uniform. A flatter heatmap means you can move your head or your chair without the bass response changing drastically.

Step 4: Check the RT60 Improvement

The Simulation provides a calculated % Improvement.

- **The Target:** Aim for the Controlled range (0.25–0.55 s).
- If your room starts at 1.20 s at 40 Hz, adding four Nodal 40 units can typically drive that down by 35–50%, bringing you into the professional reference standard.